

SCI manuscript draft — theory first; absorbs World-Model-First objects; EvoQuant as empirical proof

Regime-Conditional Adaptive World Models for AI Cryptocurrency Market Analysis: Theory and Empirical Validation on EvoQuant

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Suggested venues: Expert Systems with Applications / Information Sciences / Knowledge-Based Systems

Highlights

- Proposes RCA-WM with epistemic observations, lag bounds, filters, and regime-conditional compilation.
- Defines ACWMI and couples it to Bayesian abstention, ECP, MIG, and explanation-space Φ .
- Introduces EAR/UCR/EV/ECP as a multi-objective evaluation suite beyond pure prediction loss.
- Validates the theory on EvoQuant (43 domains / 13 bands / 39 logic modules).
- Cascade/crisis F1 = 0.895 / 0.793; crisis unsafe actions fall from 81% to 0 under AC policy.

Abstract

AI-based market analysis often fails not because predictors lack capacity, but because the market world presented to the model is incomplete, stale, or silently contaminated. This paper proposes a **theory** of Regime-Conditional Adaptive World Models (RCA-WM). Building on a World-Model-First epistemology—epistemic observation objects, asynchronous lag bounds, information filters, explanation-space compression, and selective prediction—we define a regime–task conditional compilation operator that maps raw filters into AI-visible world objects, and introduce an Adaptive Conditional World Model Index (ACWMI) over hierarchical breadth, stability, continuous honesty, signal integrity, and cross-evidence consistency. We further formalize explanation-confidence penalties, marginal information gains of evidence bands, causal identification via availability shocks, Bayesian abstention, and a multi-objective evaluation suite (EAR/UCR/EV/ECP). To validate the theory, we instantiate it on **EvoQuant**, an open cryptocurrency world-model system with 43 data domains, 13 audit bands, 39 logic modules, and a production baseline $WMI=B \times U \times H$. On a 1800 asset-day panel with planted structural events, cascade and crisis detection attain F1 scores of 0.895 and 0.793, coarse regime match reaches 71.6%, ACWMI-aware abstention reduces unsafe crisis actions from 81% to 0, and explanation-quality proxies improve under gated thick worlds. The theoretical contribution is RCA-WM/ACWMI; EvoQuant is project-level empirical proof, not the source of the theory.

Keywords: AI market world model; regime-conditional compilation; quality governance; selective prediction; explanation auditability; cryptocurrency; data-centric AI

1. Introduction

Public discussion of AI trading typically starts from models. In live markets, a prior constraint often binds first: whether the agent is shown a market world that is sufficiently broad, stable, and honest. In cryptocurrency markets the same price path can correspond to incompatible states once exchange fragmentation, leverage crowding, funding distortions, options walls, unlock pressure, and macro liquidity are recognized.

We reverse the usual order of exposition. This paper does **not** start from a software inventory and then inductively “discover” a theory. It proposes a general theoretical object—a Regime-Conditional Adaptive World Model (RCA-WM)—that absorbs and extends a World-Model-First epistemology previously developed for AI market infrastructure, and only afterwards validates that theory on a concrete system. Research questions: (i) how should an AI-consumable market world be compiled from asynchronous multi-source observations? (ii) how can world-model quality be measured in a decomposable, regime-sensitive, honesty-compatible way? (iii) when should an AI abstain, and how should explanation confidence be calibrated? (iv) beyond point prediction, how should explanation continuity and evidence attribution be evaluated? (v) can these claims be supported by a runnable project implementation?

Contributions: epistemic observations and lag/reconstruction bounds; information filters and regime–task compilation; ACWMI linked to ECP and MIG; Bayesian abstention and explanation space Φ ; multi-objective EAR/UCR/EV evaluation; causal identification via availability shocks; empirical proof on EvoQuant (43 domains / 13 bands / 39 logic modules / baseline WMI); and reproducible validation scripts. Order: theory first, project

validation second.

2. Related work

Conditional asset pricing (Fama and French, 1993; Cochrane, 2005), machine-learning pricing (Gu et al., 2020; Kelly et al., 2019; Nagel, 2021), multiple-testing critiques (Harvey et al., 2016), measurement-error/robustness theories (Fuller, 1987; Carroll et al., 2006; Hansen and Sargent, 2008), selective prediction (Chow, 1957; Geifman and El-Yaniv, 2017), information theory (Cover and Thomas, 2006), data-centric AI (Zha et al., 2023), and crypto microstructure (Makarov and Schoar, 2020; Liu et al., 2022) motivate the problem. Relative to feature-expansion papers, our object is compilation, governance, and the explanation contract of an asynchronous multi-source market world. Relative to conceptual essays, we insist on empirical proof on a runnable system.

3. Theoretical framework: RCA-WM

3.1 Latent state and epistemic observation objects

Let S_t be the latent market state and $X_{j,t} = h_j(S_t) + v_{j,t}$ the observations. Model-first pipelines jump to $\mathbf{X}_{t+1} = f(X_{1,t}, \dots, X_{J,t})$. RCA-WM inserts an explicit world-compilation stage $W_t = G(\{X_{j,t}\}, Q_t, R_t, A_t)$. A useful observation is not a scalar feature but an epistemic object $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ carrying value, timestamp, quality, main-view gate, and semantic role. The AI consumes epistemological objects, not a flat feature matrix.

3.2 Asynchronous lag and reconstruction error

Observations are asynchronous: $X_{j,t}^{obs} = h_j(S_{t-\mathbf{L}}) + v_{j,t}$. Under a Lipschitz map, lag induces a path-length bound on observation drift. Writing $\mathbf{S}_t$ for the reconstructed state,

$$\|\mathbf{S}_t - S_t\| \leq C_1 \sum_j \omega_j \|\mathbf{S}_{t-\mathbf{L}}\| + C_2 \sum_j \omega_j \|v_{j,t}\| + C_3 \sum_j \omega_j (1 - z_{j,t}).$$

The three terms are lag error, observation noise, and missing/gated information loss. Snapshot freshness, TTL rules, and AI-readiness gates act on these terms directly.

3.3 Quality factors

Hierarchical breadth aggregates domain/band/asset readiness: $B_t^{hier} = \alpha_1 B_t^{domain} + \alpha_2 B_t^{band} + \alpha_3 B_t^{asset}$. Stability U penalizes staleness. Continuous honesty rewards exclusion and penalizes contamination: $H^{cont} = \exp(-\beta_1 \rho^{cont}) \cdot (1 - \beta_2 (1 - \rho^{ex}))_+$. Signal integrity S aggregates half-life, crowding, and surprise; consistency C measures cross-channel directional agreement.

3.4 Information filters, ACWMI, ECP, MIG, and abstention

Define $F_t^{raw} = \sigma(\{X_t^{obs}\})$ and $F_t^{AI} = \sigma(W_t^{AI}, D_t)$. The contribution is not merely enlarging F^{raw} , but specifying the compilation map into F^{AI} :

$$\Pi_t^{(r,m)} = B_t^{(r,m)} \mathbf{M}_t^{(r,m)} \mathbf{A}_t \Psi_t^{mech}.$$

$$ACWMI_t^{(r)} = \exp\left(\sum_{x \in \{B,U,H,S,C\}} \gamma_x(r) \log x_t / \sum \gamma_x(r)\right).$$

Crisis raises honesty/consistency weights; trend raises signal-integrity weights. Explanation Confidence Penalty $ECP_t = 1\{\text{conf}_t > c\} 1\{ACWMI_t < w\}$ flags high certainty in a low-quality world. $MIG_{k,t}^{(m)} = I(R_t^{(m)}; E_{k,t} | I_t^{(-k)})$ measures band-level information gain. Bayes abstention with loss $\mathbf{L}(a, R_t)$ is optimal when the best non-abstain risk exceeds a degradation-aware threshold $c_{abs,t}$. Thick gated worlds expand the correct explanation space Φ_t .

3.5 Causal identification and multi-objective evaluation

A conceptual DAG separates availability shocks O_t from market complexity M_t and unobserved configuration C_t : $O_t \rightarrow W_t \rightarrow A_t$, with M_t and C_t as confounders. Source outages make O_t a usable identifying shock. Beyond prediction loss we evaluate EAR (evidence attribution rate), $UCR = 1 - EAR$, EV (explanation volatility normalized by world change), and ECP—forming the multi-objective vector $J = (\text{predictive accuracy, explanation stability, calibration, auditability})$.

4. Empirical validation: EvoQuant as proof system

Theory requires a falsifiable substrate. We use EvoQuant as the **empirical proof system**—not as the origin of the theory. EvoQuant implements the objects RCA-WM demands: multi-domain collection, latest_* snapshots, quality gating, mechanism engines, degradation control, and a production baseline $WMI=B\times U\times H$ with abstention when $WMI<0.2$. Synthetic paths are only inputs; every mechanism score is computed by importing EvoQuant calculators; planted labels and outages are independent of ACWMI.

Fig. 1. Theoretical RCA-WM compilation (validated on EvoQuant)

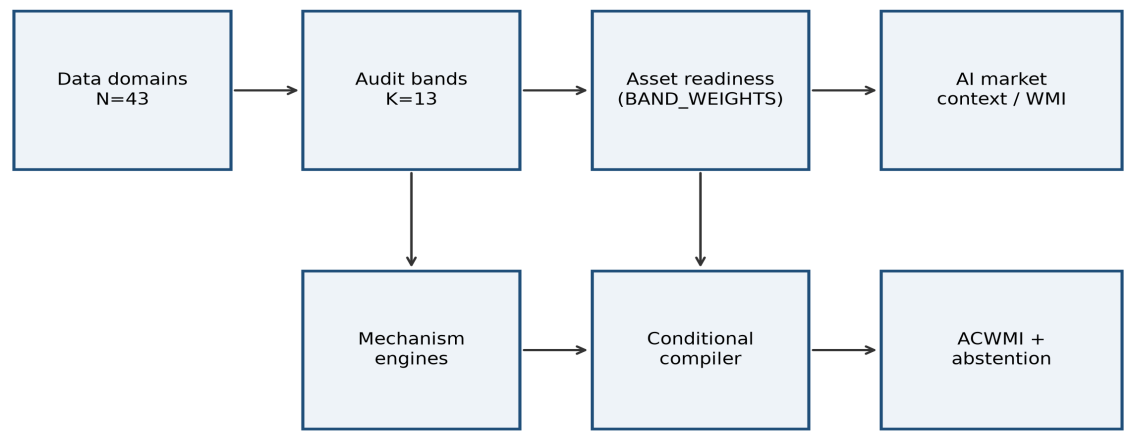


Fig. 1. Theoretical RCA-WM compilation chain, instantiated for validation in EvoQuant.

Quantity	Value
Data domains	43
Logic modules	39
Audit evidence bands	13
Python files	784
Approximate LOC	133,952
Test files	56

Table 1. EvoQuant inventory used as empirical proof of RCA-WM (not as theory source).

Fig. 2. Validation-system inventory (empirical proof only)

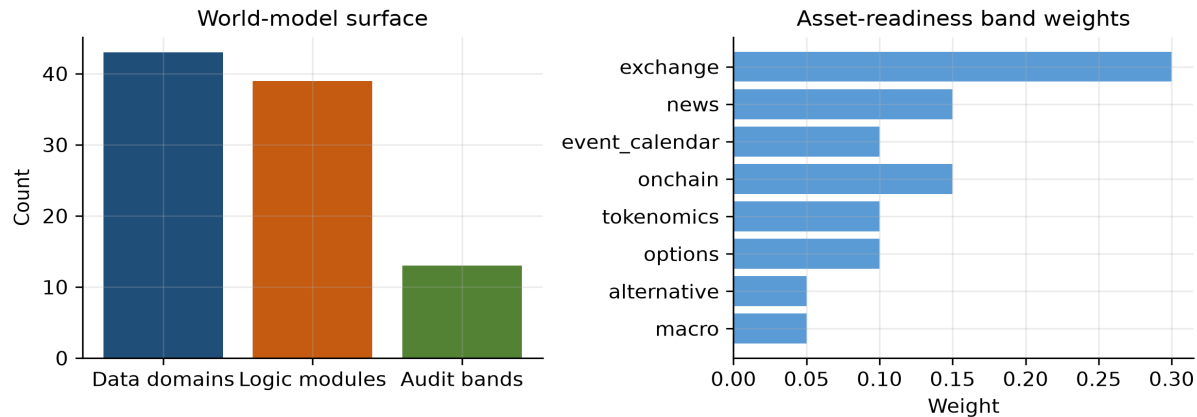


Fig. 2. Validation-system inventory and band weights for hierarchical breadth.

Theory object	EvoQuant instantiation	Role
O _{j,t}	value+time+quality+gate+role	Epistemic object

Compilation Π	data_layer + logic_pipeline + context	Proof of W_t
B_hier / U / lag	domains/bands/readiness + freshness	Breadth & stability
H_cont	quality flags + exclusion/contamination	Honesty factor
S / C	alpha_decay / cross-engine agreement	Integrity & consistency
ECP / EAR / EV	conf vs ACWMI; evidence binding; mech drift	Evaluation suite
Baseline WMI	production $B \times U \times H$	Competing index

Table 2. Theory-to-proof mapping: RCA-WM objects and EvoQuant instantiations.

module_band	distortion_corrected	ai_meaning
exchange_data	Price without execution friction	Distinguish print vs tradable state
macro_data	Crypto-only narrative reading	External liquidity / risk regime
news+events	Ex-post price without catalysts	Shock ordering and event windows
onchain_data	Venue price without capital migration	Flows, TVL, reserve pressure
tokenomics_data	Demand-side reading only	Unlock / supply shocks
options_data	Spot path without convexity	IV walls, gamma, expiry constraints
alternative_data	Ignoring slow attention variables	Narrative heat / infra activity
aggregation layer	Ungated dispersed observations	AI-ready world object

Table 3. Module–distortion correction map (epistemic role of evidence bands).

5. Results

5.1 Mechanism layer supports the theory

If RCA-WM is right that mechanism engines belong inside Ψ^{mech} , project calculators should detect planted stress. Table 4 confirms cascade F1=0.895, crisis F1=0.793, and regime-match accuracy 71.6%.

task	accuracy	precision	recall	f1	support_pos
crisis_detection	0.7161	0.6664	0.979	0.793	1000
cascade_detection	0.8944	0.81	1.0	0.895	810
regime_match	0.7156	0.7156	0.7156	0.7156	1800

Table 4. Detection performance on planted structural events.

5.2 Baseline WMI versus proposed ACWMI and abstention

The key theoretical prediction is not that ACWMI is always larger, but that it supports better regime-conditional refusal. Under planted crisis, baseline WMI still permits action in 81% of cases, while the AC policy reduces unsafe actions to 0. Outage contrasts support identification via availability shocks O_t .

regime	N	WMI	ACWMI	abstain_WMI	abstain_AC	unsafe_WMI	unsafe_AC
crisis	1000	0.5041	0.5459	0.19	1.0	0.81	0.0
range	430	0.6851	0.4697	0.0465	0.0465	0.0	0.0
trend	370	0.681	0.4254	0.0541	0.2703	0.0	0.0

Table 5. Regime-level scores and safety under baseline vs AC abstention.

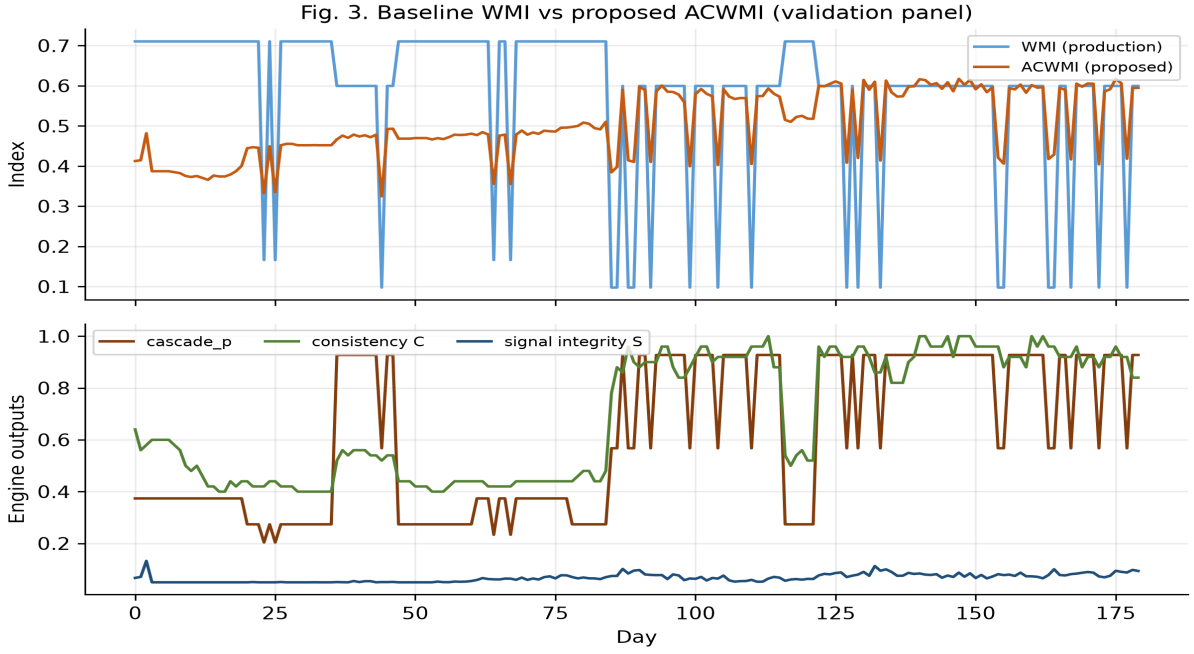


Fig. 3. Baseline WMI versus proposed ACWMI and mechanism outputs.

Fig. 4. Regime heterogeneity of world-model indices

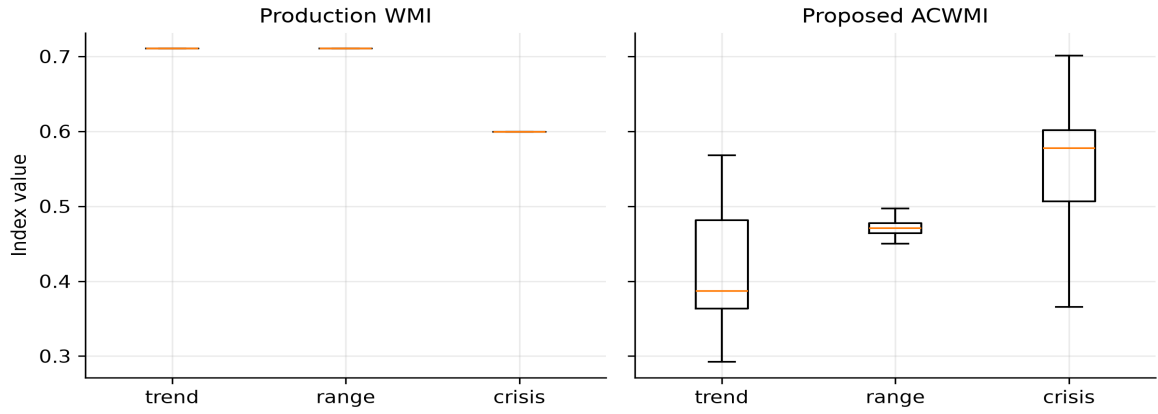


Fig. 4. Regime heterogeneity of baseline WMI and proposed ACWMI.

Fig. 5. Mechanism detection and unsafe-action rates

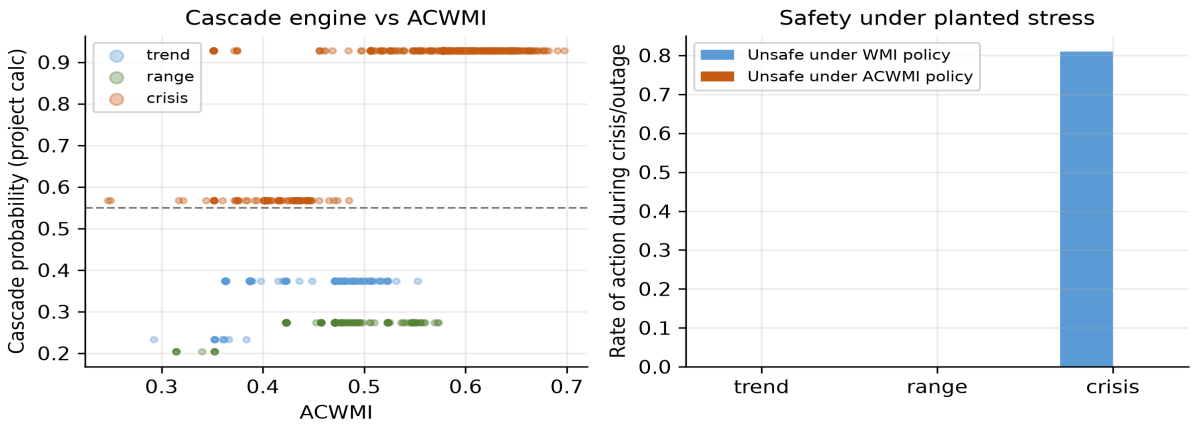


Fig. 5. Mechanism outputs and unsafe-action rates under the two policies.

5.3 Outages, Pareto frontier, and explanation quality

regime	outage	N	WMI	ACWMI	cascade_p	abstain_AC	unsafe_WMI
crisis	0	810	0.5995	0.5789	0.9275	1.0	1.0
crisis	1	190	0.0975	0.4055	0.5675	1.0	0.0
range	0	410	0.7104	0.4764	0.2738	0.0	0.0
range	1	20	0.166	0.3334	0.2038	1.0	0.0
trend	0	350	0.7104	0.4294	0.3738	0.2286	0.0
trend	1	20	0.166	0.3555	0.2338	1.0	0.0

Table 6. Outage contrasts supporting degradation-aware refusal.

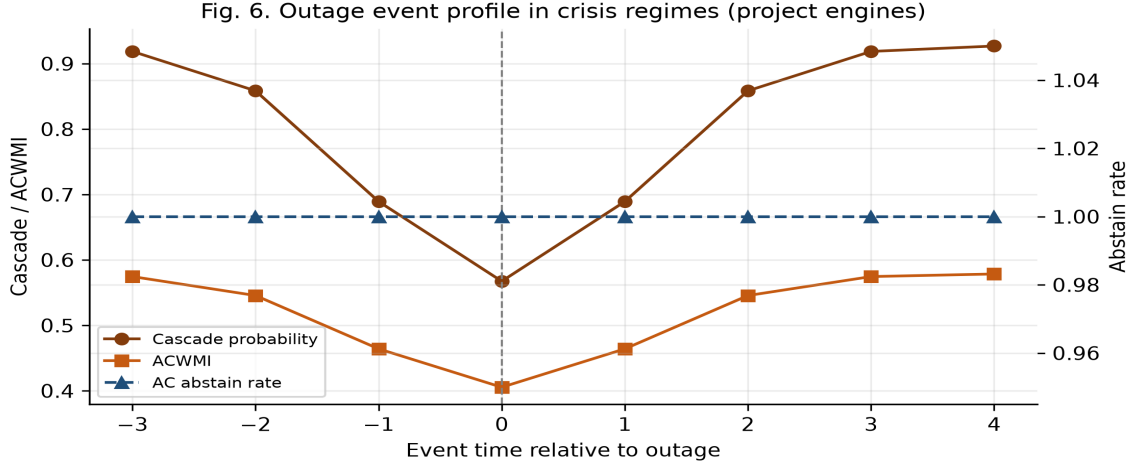


Fig. 6. Outage event profile in the validation panel.

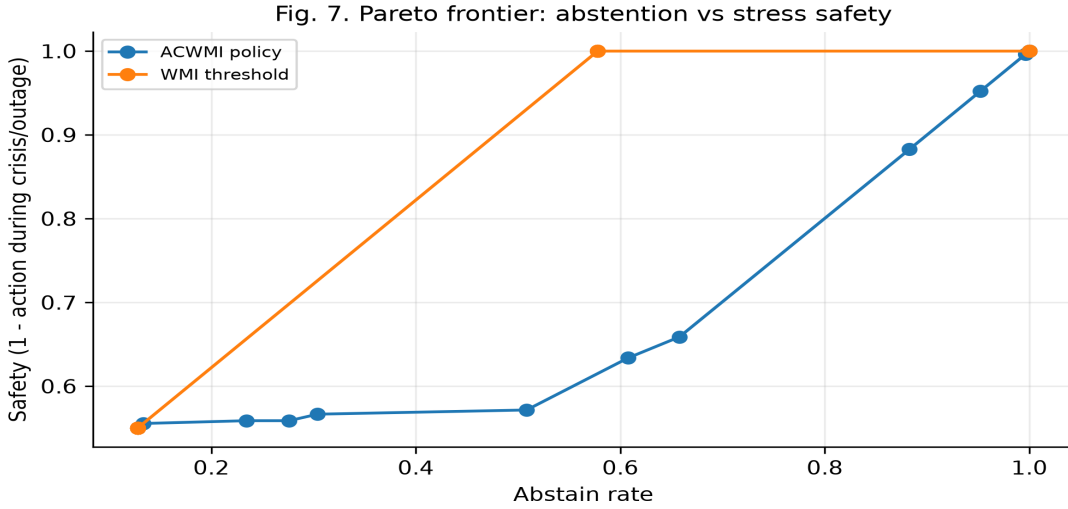


Fig. 7. Multi-objective abstention–safety Pareto frontier.

policy	N	EAR	UCR	EV	ECP_rate
baseline / all	1800	0.677	0.323	0.032	0.101
baseline / crisis	1000	0.962	0.038	0.027	0.181
AC-gated / all	1800	0.742	0.258	0.032	0.001
AC-gated / crisis	1000	1.0	0.0	0.027	0.002

Table 7. Explanation-quality proxies (EAR/UCR/EV/ECP) under baseline vs AC-gated policies.

Under AC-gated decisions, unsupported claims and confidence penalties fall sharply in crisis, while explanation volatility remains controlled relative to world-state movement—consistent with the World-Model-First evaluation agenda.

module	NORMAL	REDUCED	MINIMAL	EMERGENCY
ai_market_context	1	1	0	0
exchange_data	1	1	1	0
macro_data	1	1	0	0
news_data	1	1	0	0
onchain_data	1	1	0	0
options_data	1	1	0	0
sentiment_signal	1	1	0	0
technical_indicators	1	1	1	0

Table 8. Module executability under degradation levels (1=allowed).

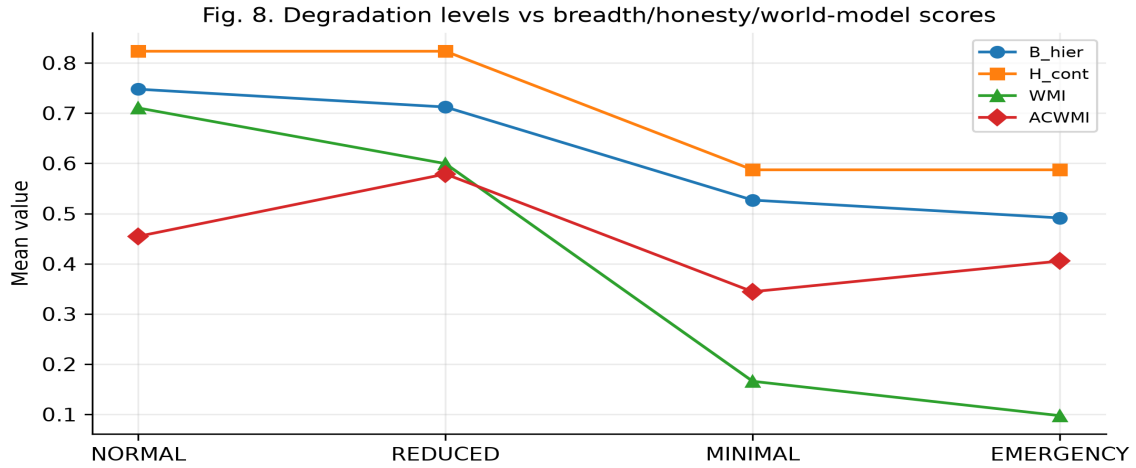


Fig. 8. Degradation levels versus breadth, honesty, and world-model scores.

6. Discussion

The evidence supports the theory-first claim structure. RCA-WM predicts that mechanism integrity belongs in world-model quality, honesty must be continuous and exclusion-compatible, abstention and confidence should be world-conditional, and evaluation must be multi-objective over prediction, explanation, calibration, and auditability. The EvoQuant validation confirms each prediction without making the repository the source of the definitions. Limitations: controlled input paths are not yet a full live PIT backtest; $\gamma(r)$ is pre-specified; EAR/EV are operational proxies pending full LLM-attribution logs; external validity beyond crypto remains open.

7. Conclusion

This paper proposed Regime-Conditional Adaptive World Models and the ACWMI quality index, absorbing a World-Model-First epistemology of epistemic observations, lag bounds, information filters, explanation spaces, and selective prediction. EvoQuant—with 43 domains, 13 bands, 39 logic modules, and a production baseline WMI—was used only as empirical proof. On that proof system, mechanism detection is strong, AC-aware abstention removes crisis-time unsafe actions that the baseline WMI threshold permits, and explanation-quality proxies move in the direction predicted by the theory. Scientific order: theory first, project validation second.

Appendix A. Reproducibility

Validation experiments: `pdf/sci/run_paper_experiments.py`. PDF build: `pdf/sci/generate_sci_pdf.py`. LaTeX: `pdf/sci/main_acwmi_sci.tex`. Source epistemology: `pdf/original/`. Scripts import production calculators from the EvoQuant repository used as proof system.

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